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from pyspark.sql import SparkSession
from pyspark.sql.functions import col

# Create Spark Session
spark = SparkSession.builder.appName("EchoChain_Member2").getOrCreate()

# Load Dataset
df = spark.read.csv("EchoChain_1000_Rows_Dataset.csv", header=True, inferSchema=True)

print("Original Dataset")
df.show(5)

# Bronze Layer
bronze_df = df
bronze_df.write.mode("overwrite").parquet("Bronze")

# Silver Layer
silver_df = bronze_df.dropDuplicates()

for column in silver_df.columns:
    silver_df = silver_df.fillna({column: "Unknown"})

silver_df = silver_df.toDF(*[c.lower().replace(" ", "_") for c in silver_df.columns])
silver_df.write.mode("overwrite").parquet("Silver")

# Gold Layer
gold_df = silver_df

gold_df.write.mode("overwrite").parquet("Gold")

print("Cleaned Dataset")
gold_df.show(5)

print("Pipeline Completed Successfully")
```